
RealBlend – User Manual & Documentation

Version: 1.1

 **Support:** winnerwaffle@gmail.com

 **Tutorial Video:** <https://youtu.be/qQ-P3XAelWU>

I strongly recommend watching the above video

Introduction

RealBlend is a complete mesh editing solution for Unity.
It combines:

-  Multi-layer vertex painting
-  Intuitive sculpting tools
-  Procedural mesh creation

All inside one powerful Editor window.

Key Features

5-Layer Vertex Painting

Paint up to **5 material layers** using:

- RGBA channels
- Plus a Base layer

Mesh Sculpting

- Push & pull geometry
- Smooth surfaces
- Refine topology

- Edit meshes directly inside Unity

Procedural Mesh Creation

Generate:

- Floors
- Walls
- Curved surfaces

Instantly paint-ready.

Smart Data Storage

Uses a **non-destructive workflow** via the `VertexPaintStorage` component.

Mask Packer

Built-in utility to pack HDRP/URP texture masks correctly.

2 Installation & Setup

Follow these steps to get started:

Import the Package

- Import via **Unity Package Manager**
OR
 - Double-click the `.unitypackage` file
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Define Your Render Pipeline

RealBlend works with:

- Built-in
- URP

- HDRP

⚠ Important: Your shader must support **Vertex Colors**.

Access the Tools

Open via:

Tools → **RealBlend** → **Real Blend**

Mask Packer is located in the same menu.

Scene Preparation

MeshCollider Required

Any object you want to paint or sculpt must have a **MeshCollider**.

VertexPaintStorage Component

- Required to save edits
- The tool will prompt you to add it automatically

Compatible Material

Your object must use:

- RealBlendBuiltInVariation shader
OR
 - Included URP/HDRP Shader Graphs
OR
 - Any shader that reads Vertex Colors
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Paint Tab

The core texturing interface.

Two workflow modes available:

3.1 Workflow Modes

LayeredBlend

Used for multi-texture blending:

- Base layer
- Layer 1
- Layer 2
- Layer 3
- Layer 4

VertexColor

Standard color painting with palette support.

3.2 Brush Controls

Brush Settings

- Size
- Opacity
- Hardness
- Mouse Sensitivity
- Brush Alpha (stamp textures)
- Brush Rotation

Lock Selection

Prevents accidental object switching while painting.

Flood Fill

Fill the entire mesh with selected layer or color.

Shortcut

Hold **Shift** while painting to erase.

Sculpt Tab

Modify the physical shape of meshes.

Sculpt Modes

Move / Push / Pull

Displace vertices:

- Along normals
- Or along mouse direction

Nudge

Click-drag to slide points across the hit plane.

Flatten Height

Flatten geometry to a target height on:

- Local X
- Local Y
- Local Z

Smooth

Relax vertices and average out noise.

Refine (Add Detail)

Subdivide triangles locally to increase vertex density.

 Note: This changes mesh topology.

Remove

Delete selected triangles and unused vertices.

Revert Pose

Gradually morph vertices back to their original stored position.

Create Tab

Generate evenly distributed, paint-ready meshes.

Orientation Options

- Floor (XZ)
 - Wall (XY)
 - Curved Wall
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Curved Wall Controls

- Curvature
 - Completeness
 - Pivot Location
 - Mesh Center
 - Circle Center
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Resolution

Set **Density (Verts/m)** to control vertex density.

Higher density = more detailed sculpting & painting.

Shader Integration & Variables

Includes:

- `PolybrushMixer.cginc`
 - Pre-made shaders for Built-in, URP, HDRP
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6.1 Channel Mapping (GBAR Workflow)

Layer	Channel	Usage
Base	Background	Underlying texture
Layer 1	Green (G)	Layer 1 weight
Layer 2	Blue (B)	Layer 2 weight
Layer 3	Alpha (A)	Wetness mask
Layer 4	Red (R)	Variation mask (Set A vs Set B)

6.2 Key Shader Variables

`_BlendNoise`

Adds breakup to reduce procedural-looking edges.

`_L12_BlendBridge`

Helps Layer 1 and 2 transition smoothly.

`_LayerContrast`

Controls hardness of height-based blending.

Variation (Set B)

Uses Red vertex channel to blend between:

- `_Tint_B`

- `_Saturation_B` and Set A values.
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Data Management & Baking

7.1 VertexPaintStorage

This component acts as the **Save File** for your mesh.

It stores:

- Vertex Colors
- Vertex Positions
- UVs
- Triangles

Ensures edits are not lost when closing scenes.

7.2 Baking (For Builds)

To make changes permanent:

1. Select the object
2. In `VertexPaintStorage`
3. Click **Bake to Asset (Permanent)**

✓ Creates a mesh file inside:

`Assets/BakedVertexPaintMeshes`

After baking:

- You can safely remove the component.
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Mask Packer Utility

Located at:

Tools → RealBlend → Mask Packer

Packing Format

Channel	Usage
R	Metallic
G	Ambient Occlusion
B	Roughness
A	Height Mask

Built-in Pipeline Warning

The current built in shader is missing POM features and requires greater effort of tuning the "Contrast" variable for adequate results.

Summary

RealBlend gives you:

-  Professional multi-layer vertex painting
-  Advanced sculpting inside Unity
-  Procedural mesh generation
-  Non-destructive workflow
-  Pipeline-compatible shaders

All designed to speed up environment creation and give you full artistic control.
